

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

SA7 – Investment and Finance

May 2025

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q. 1) Topic: *Investment Management, Capital Management, and Investment Strategy (Risk-Based Capital, Asset-Liability Matching)*

i) Product Liability Characteristics

Syllabus Topic: *Meeting investor requirements, including investment strategies to meet liabilities.*

- With-Profit Policies (Participating Business): These policies promise bonuses that depend on investment returns. Investment characteristics include a balanced mix of low-risk bonds and higher-risk equities to generate returns while ensuring sufficient capital to meet policyholder guarantees.
- Non-Profit Policies: Typically, these policies involve fixed premiums and benefits. Investment characteristics focus on low-risk investments (such as government bonds) to match the predictable liabilities.
- Unit-Linked Policies (ULIPs): These are investment-linked insurance products where policyholders bear the investment risk. The asset mix depends on the investor's choice but should include equities, bonds, and other instruments for potential growth.
- Annuity Contracts: These guarantee regular income for policyholders, so the investment strategy is designed to match long-term liabilities with low-risk bonds, minimizing exposure to market volatility.

(5)

ii) Impact of Inflation and Interest Rates

Syllabus Topic: *The framework for investment management, including corporate finance principles.*

- Impact on Liabilities: Inflation increases the present value of liabilities, especially for products like annuities with inflation-linked payouts. Rising interest rates may increase the cost of managing long-term liabilities.
- Impact on Assets: Inflation reduces the real return on fixed-income assets, while rising rates can lead to price declines in long-duration bonds.
- Solvency Concerns: The company's solvency could be at risk if the liabilities increase faster than assets, especially if investments are mainly in fixed-income assets.
- Investment Strategy: Higher interest rates could make government bonds more attractive but may hurt long-term growth if equity allocations are reduced.
- Policy-holder expectations: Participating and ULIP customers expect competitive returns; failing to match peer performance risks lapses and reputational damage.

(5)

iii) Capital Raising Options

Syllabus Topic: *The framework for investment management, with an emphasis on corporate finance.*

- New Equity Capital:
 - Merits:
 1. Directly boosts capital to meet RBC requirements.
 2. Improves solvency and provides financial stability.
 3. No repayment obligations compared to debt.
 - Demerits:
 1. Dilution of ownership and control.
 2. Potentially lowers earnings per share (EPS) due to increased share count.
 3. May negatively affect stock price if market perceives dilution negatively.
- Subordinated Debt:
 - Merits:
 1. Lower cost of capital than equity.
 2. No dilution of ownership or control.
 3. May provide tax benefits since interest payments are tax-deductible.
 - Demerits:
 1. Repayment obligations and interest expenses.
 2. Increases leverage and financial risk.
 3. May be difficult to issue if market conditions are poor.

(8)

iv) Higher-Yield Investment Strategy

Syllabus Topic: *Management and risk control for an investment manager, including portfolio management.*

- Advantages:
 1. Higher Returns: Potential for higher long-term returns through risky assets like equities and high-yield bonds.
 2. Increased Surplus: Higher returns lead to greater surplus, which can be used for policyholder bonuses or capital growth.
 3. Capital Diversification: Increases portfolio diversification by including more asset classes.
 4. Inflation Hedge: Equities and infrastructure assets often outperform in inflationary environments.
 5. Liquidity for Growth: Offers liquidity to take advantage of growth opportunities.
 6. Regulatory Arbitrage: Can potentially improve risk-based capital by allocating to assets that are less capital-intensive under RBC.
 7. higher expected return, diversification, potential inflation hedge, opportunity to buy mis-priced credit, tactical use of IRDAI “Other-Investments” bucket.
- Risks:
 1. Higher Capital Charges: The RBC framework assigns higher capital requirements to riskier assets like equities and lower-rated bonds.
 2. Volatility: Greater asset volatility increases the risk of not meeting policyholder guarantees.
 3. Liquidity Risk: Riskier assets like private equity or real estate can have lower liquidity, posing risks during market stress.
 4. Market Risk: Increased exposure to market downturns can significantly affect the company’s financial stability.
 5. Risk to Policyholder Expectations: If riskier investments underperform, it could lead to lower than expected returns for policyholders, damaging the company’s reputation.
 6. credit-risk (default and downgrade), higher RBC charges (e.g. 17 %–22 % for listed equity, 12 %-15 % for BB/BBB credit), liquidity risk for real estate/infra funds, concentration limits (e.g. 15 % per issuer group, 10 % unlisted equity), governance capacity, pro-cyclicality.

(10)

v) De-risking and Asset–Liability Matching

Syllabus Topic: *Meeting investor requirements and risk management strategies.*

- Advantages of De-risking:
 1. Guarantee Security: Ensures that the portfolio can meet future liability payments, especially for guaranteed products.
 2. Solvency Stability: Reduced volatility in the asset portfolio leads to greater solvency stability.
 3. Interest Rate Matching: Long-term bonds match the duration of liabilities, helping to mitigate interest rate risk.
 4. Regulatory Compliance: Aligns with regulatory requirements for matching assets and liabilities.
 5. Reduced Market Sensitivity: Less exposure to equity markets lowers the likelihood of capital shortfalls during market downturns.
- Drawbacks of De-risking:
 1. Lower Returns: The portfolio may generate lower returns, especially if government bonds underperform.
 2. Potential Missed Opportunities: A conservative strategy may miss higher returns from growth assets.
 3. Capital Inefficiency: Holding low-yielding bonds could be less efficient from a capital perspective under the RBC framework.

(7)

vi) Recommendation

Syllabus Topic: *Solving problems related to overall financial management of portfolios.*

Strategic overview. ABC Life should pursue a dual-track approach: (a) shore up regulatory capital ahead of the 2026 RBC go-live, and (b) pivot the asset mix so that guaranteed liabilities are tightly hedged while surplus assets continue to earn a competitive return for participating and ULIP policy-holders.

Raise hybrid capital. Issue INR-denominated Tier II subordinated debt up to 25 % of available capital and launch a rights-issue for any residual gap so that post-issue coverage sits at least 300 bps above the projected RBC minimum.

Optimise cost of capital. Debt coupons are tax-deductible and avoid ownership dilution; the equity tranche is deliberately modest to protect earnings per share while still signalling commitment to solvency.

Segregate the portfolio into two buckets. A guarantee-backing portfolio covers with-profit guarantees and annuities, while a surplus-growth portfolio supports bonuses and ULIP fund choices.

Guarantee-backing asset mix. Allocate 80 % to long-dated G-Secs/SDL STRIPS, 15 % to AAA PSU bonds and 5 % to payer swaptions that cap reinvestment risk if yields fall.

Ensure duration and cash-flow matching. The weighted-average duration of backing assets should be within ± 0.5 years of liability duration; cash-flow testing confirms 95 % of benefits are covered by contractual coupons.

Surplus-growth asset mix. Invest 45 % in diversified equities (20 % global ETFs to reduce home bias), 25 % in infrastructure debt funds/InvITs, 10 % in REITs and 20 % in high-grade corporate bonds that carry lower RBC charges than equities.

Inflation resilience. Within the equity sleeve, tilt 10 % towards consumer-staples and energy companies that historically outperform during persistent inflation.

Dynamic rebalancing triggers. If RBC coverage falls below 155 %, automatically shift 5 % of risky assets into long bonds; when coverage exceeds 175 %, reverse the move. This rules-based overlay keeps solvency volatility in check.

Liquidity management. Maintain a 5 % buffer in T-Bills and triparty repos to fund derivative margin or meet large ULIP switches within T+1.

Currency and rate hedging toolkit. Use INR interest-rate swaps to fine-tune duration, swaptions for tail protection and USD/INR forwards to neutralise any overseas equity exposure.

Robust credit-risk governance. Set single-issuer limits at 2 % of portfolio market value and ensure aggregate exposure to BBB credit stays below 10 %, comfortably within IRDAI 'Other-Investments' caps.

Comprehensive stress testing. Perform quarterly RBC-consistent shocks – e.g., 200 bps parallel yield shift, 30 % equity fall and BBB-to-BB downgrades – and escalate breaches to the Board Risk Committee.

Transparent policy-holder communication. Update the bonus philosophy and ULIP fund literature so that customers understand the new mix and its risk/return implications, thereby managing lapse risk.

Operational readiness. Upgrade middle-office systems for daily RBC monitoring, collateral calls and covenant reporting on the Tier II debt; provide staff training on the new workflows.

Annual strategic review clause. Commit to a full ALM and capital review every year, or ad-hoc if the 10-year G-Sec yield moves by more than 75 bps in a single quarter, ensuring the strategy remains fit for purpose.

By combining fresh hybrid capital with a disciplined, liability-aware asset strategy, ABC Life can comfortably exceed the forthcoming RBC hurdle, safeguard guarantees and still generate the investment upside that policy-holders expect.

(15)

[50 Marks]**Q. 2)****i) Global Trade Tensions and the USD/INR Exchange Rate**

Syllabus Reference: *The framework for investment management, including global influences on capital markets.*

Exchange-rate pressure from global trade frictions

1. Safe-haven dollar demand – Tariffs and “on-shoring” rhetoric prompt global investors to buy USD assets; INR tends to weaken or stay volatile.
2. Wider Indian trade deficit – Costlier imported inputs (oil, electronics) enlarge the current-account gap, adding structural INR pressure.
3. Higher U.S. real yields – Fed policy encourages EM outflows into Treasuries; passive global-bond funds trim India allocations.
4. Risk-off sentiment – EM equity funds reduce gross exposure; India sees episodic FII pull-outs when VIX spikes.
5. Swap-hedge costs rise – Wider USD funding spreads lift the cost of rupee hedges; this lowers India’s risk-adjusted equity appeal.
6. Sector effects – Export-heavy IT stocks may benefit from INR weakness, but Financials suffer as wholesale-funding costs jump.
7. Regulatory frictions – Stricter FPI KYC/UBO rules slow approval pipelines; some offshore funds hold back.
8. Bond-index offset – Prospective inclusion of India in global bond indices could partly counter equity outflows, but timing is uncertain.
9. RBI intervention limits – Spot and forward-book sales can smooth, not reverse, INR moves; large FPI exits still move the tape.
10. Net result – Wider USD/INR trading band, thinner FII flows, and a higher equity-risk premium for India

(10)

ii) Impact of INR Depreciation on the Bond Portfolio

Syllabus Reference: *Investment strategies to meet liabilities and risk control for investment managers.*

Currency risk on the USD bond portfolio

A. INR valuation impact

Scenario	FX rate (₹/US \$)	Portfolio value (₹ bn)	Change vs base
Today	85.00	21.25	–
INR appreciates 6 %	79.90	19.98	- 1.27 bn
INR depreciates 6 %	90.10	22.53	+ 1.28 bn

(US \$ 250 m × FX ÷ 1,000)

B. Two hedging actions

Hedge	One key benefit	One key drawback
Rolling 3-month USD/INR forwards	Low-cost, linear hedge; easy to size exactly to bond principal & coupons	Negative carry when INR forward premium is wide; frequent rollover risk

<i>Hedge</i>	<i>One key benefit</i>	<i>One key drawback</i>
<i>12-month USD put / INR call options</i>	<i>Caps downside but keeps upside if INR depreciates less than expected</i>	<i>Premium outlay; option market depth thin for sizes > \$1 bn</i>

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iii) Tactical Rebalancing in Response to Sector-Specific Risk

Syllabus Reference: *Management and risk control for investment managers, including portfolio management.*

. Loss estimate

$40\% \times (-12\%) = -4.8\%$ hit to the total equity sleeve.

B.

Set a 5 % NAV draw-down trigger and pre-clear the action list with the Investment Committee.

Pre-defined governance avoids ad-hoc decisions under stress.

Use sector-index futures (short NIFTY-Bank) or total-return swaps to obtain instant Financials-beta reduction while cash trades are scheduled.

Derivative overlay slashes exposure on day 1 without large-block selling.

Phase down the Financials cash weight from 40 % to 25 % via (Volume-Weighted Average Price algorithms and crossing networks minimises market-impact and brokerage.

Stagger sales over multiple tax years or utilise index-future “rolls” to defer crystallising capital-gains tax on the legacy book.

Redeploy 10 % into Consumer-Staples ETFs and 5 % into Healthcare stocks with low earnings cyclicality defensive tilt dampens portfolio volatility.

Fund 1 %–2 % of notional in 3-month NIFTY-50 put spreads to cap downside through the execution window

option premium is a known, limited cost.

Maintain liquidity buffers: post-trade, hold $\geq 10\%$ of AUM in cash/T-Bills and ensure any single-day turnover $\leq 20\%$ of 30-day ADTV, meeting SEBI’s liquidity-risk guidelines.

Track total transaction cost (brokerage + bid-ask + slippage) and keep it below 40 bps round-trip; back-test VaR and Stress VaR daily to confirm the new mix stays within the fund’s 95 % VaR ceiling.

(8)

[25 Marks]

Q.3)

i) Relevant Syllabus Objectives:

- Primary: 2.1, 2.1.1 (specifically for Indian Pension Fund context), 3.1.5, 3.1.3, 3.1.4, 4.1 series.
- Secondary: 1.1 (Indian assets/derivatives), 1.2 (Indian economic/market influences ¹), 1.3 (Indian legislative/regulatory framework ²), 3.2.1/3.2.2.

Objectives, Constraints, and Sponsor Covenant

- Objectives:
 - Primary Objective: Ensure all member benefits are paid securely as they fall due
 - Achieve this with a very high degree of certainty
 - Maintain stability in the funding level on the actuarial basis
 - Minimise the risk of deficits emerging in the fund
 - Adhere to regulatory guidelines for Exempt PF Trusts in India
 - Fulfil fiduciary duties under the Indian Trusts Act, 1882
- Constraints:
 - Liability Profile: Mature, closed scheme dictates focus on pensioners/deferreds

- Liability Profile: Liabilities likely long-duration, requiring long-term asset matching
- Regulatory Investment Pattern: Need to operate within permissible investment patterns (implicit or explicit) for Indian PF trusts
- Liquidity Needs: Must hold sufficient liquid assets for benefit outgo
- Indian Market Context: Prevailing G-Sec yields & availability of specific bonds constrain matching
- Sponsor Covenant Relevance:
 - Provides a backstop against deficits, influencing risk capacity
 - *However*, for a mature scheme, focus remains on asset-liability matching, reducing reliance on the covenant over time

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ii) LDI Principles and Implementation Approaches

- LDI Principles:
 - Manage assets *relative* to liabilities, not just asset benchmarks
 - Aim to reduce funding ratio volatility
 - Match asset sensitivity to liability sensitivity for key risks
 - Key risks for this Indian trust: Indian interest rates (G-Sec curve)
 - Key risks: Relevant Indian inflation measures
- Implementation: Physical Bonds:
 - Mechanism: Buy physical Indian bonds (G-Secs, SDLs, Corp Bonds)
 - Mechanism: Match bond cash flows/duration to liabilities
 - Pros: Transparent and conceptually simple
 - Pros: Directly holds income-generating assets
 - Cons: Perfect matching is difficult due to bond availability
 - Cons: Can be capital intensive to achieve high duration
- Implementation: Synthetic Replication (Derivatives):
 - Mechanism: Use derivatives like INR Interest Rate Swaps (IRS)
 - Mechanism: Receive fixed / pay float to increase asset duration synthetically
 - Pros: Capital efficient for adjusting risk exposures
 - Cons: Introduces counterparty credit risk
 - Cons: Requires operational capability for collateral management

(8)

iii) Revised SAA Recommendation

- Rationale: Current 60% equity is too high for a mature, closed Indian PF trust need to shift significantly towards matching assets
- Matching Portfolio (Target 75-85%):
 - Core Bonds (50-60%): Primarily G-Secs and SDLs
 - Core Bonds: Include high-quality (AAA/AA) Indian PSU/Corporate bonds for yield
 - LDI Derivatives Overlay: Use INR IRS to fine-tune duration hedge if needed
 - Cashflow Matching Credit (5-10%): Consider Indian InvITs for long-term cash flows
 - Cashflow Matching Credit: Requires due diligence due to complexity/liquidity risk
- Growth Portfolio (Target 15-25%):
 - Indian Equities (10-15%): Significantly reduce; focus on diversified large-caps
 - Indian Equities: Consider using low-cost Index Funds/ETFs
 - Other Diversifiers (5-10%): Maybe REITs or suitable multi-asset funds available in India
 - Other Diversifiers: AIFs likely face stricter regulatory/governance hurdles for PF trusts
- Justification: Reduces equity risk, increases liability hedge ratio balances security with modest growth potential within Indian market context introduces manageable credit/liquidity risk via non-G-Sec matching assets

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iv) Governance Considerations

- Need a clear, updated Investment Policy Statement (IPS)
- IPS should detail objectives, SAA, risk tolerance, LDI approach
- Ensure Trustees have necessary expertise or access to advice
- Provide regular training for Trustees on investment/LDI strategy

- Implement robust manager selection and monitoring processes
- Mandates must align with IPS; monitor performance and risk controls
- Maintain an integrated risk management framework
- Regular, clear reporting to Trustees on performance, risk, and compliance

(4)

[25 Marks]
